Spectral Unmixing-Guided Prompt Network for Cross-Domain Hyperspectral Image Classification

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***Abstract*—In the field of cross-domain hyperspectral image classification, single-source domain generalization (DG) frameworks are currently being explored. Most existing methods generate out-of-domain data from the source domain (SD) using techniques like style transfer, photometric, and geometric transformations. However, the domain expansion strategies designed by these methods often ignore the characteristics of mixed pixels in hyperspectral images. In addition, current hyperspectral imagery (HSI) cross-domain classification methods mainly focus on domain-invariant representations, neglecting the use of domain-specific knowledge for cooperative inference in the target domain (TD). To address these limitations, we propose the spectral unmixing-guided prompt network (SUPnet), which integrates unmixing and classification through a coordinated mechanism. This approach generates out-of-domain data based on spectral unmixing and employs a prompt learning strategy to facilitate multidomain-specific representation collaborative inference. Specifically, in order to estimate abundance values and endmembers in an unsupervised manner, a spectral unmixing generator (SUG) is designed, which constructs extended domains (EDs) by perturbing endmembers weighted by abundance values. Furthermore, a multilevel prompt learning (MPL) is developed to perform prompt learning at both the domain and endmember levels, thereby extracting specific knowledge and correlations between different domains to improve TD inference. Experimental results on five cross-domain HSI datasets covering multiple platforms (including UVA, airborne, and satellite) demonstrate that the proposed method significantly outperforms several current state-of-the-art approaches. Notably, SUG proves to be more efficient than many of advanced unmixing methods. The code will be available at: https://github.com/YuxiangZhang-BIT/IEEE_TGRS_SUPnet**

***Index Terms*—Cross domain, domain generalization (DG), hyperspectral image classification, prompt learning, spectral unmixing.**

NOMENCLATURE

HSI	Hyperspectral imagery.
DA	Domain adaptation.
DG	Domain generalization.
SD	Source domain.
TD	Target domain.
ED	Extended domain.
SUG	Spectral unmixing generator.
MPL	Multilevel prompt learning.
X and $\hat{\mathbf{X}}$	SD and ED data.
$\bar{\mathbf{M}}$ and $\hat{\mathbf{M}}$	SD and ED endmembers.
$\tilde{\mathbf{X}}_{all}$ and $\tilde{\mathbf{M}}_{all}$	Data and endmember representations of all domains.
$\mathbf{T}_{se}$	Semantic token.
$\mathbf{T}_{cls}$	Classification token.
$\mathbf{T}_{dm}$ and $\mathbf{T}_{em}$	Domain and endmember prompts.
$\tilde{\mathbf{T}}_{dm}$ and $\tilde{\mathbf{T}}_{em}$	Domain and endmember adapted prompts.
$\mathbf{a}$	Abundance value.
$\mathbf{z}$	Latent space variable.

I. INTRODUCTION

HYPERSPECTRAL imagery (HSI) is a type of remote sensing image with extremely high spectral resolution, capable of capturing continuous and finely divided spectral band information across ultraviolet, visible, near-infrared, and mid-infrared regions of the electromagnetic spectrum. The acquisition of multidimensional spatial and spectral information allows HSI to detail the physical properties of materials, thereby enabling precise identification and analysis. This capability has rendered HSI widely valuable in numerous fields such as forest resource surveys, urban environmental analysis, and biodiversity monitoring and assessment. In recent years, the cross-domain HSI classification technology, a key aspect of HSI classification, has achieved a series of remarkable advancements [1], [2], [3], [4]. This technology primarily focuses on addressing spectral shift issues between different scenes caused by factors such as sensor nonlinearities, seasonal variations, and weather conditions [5], [6], [7]. By enabling effective model transfer from the SD to the TD, it provides robust support for scene migration and accurate interpretation of hyperspectral remote sensing data.

Currently, the cross-domain HSI classification primarily encompasses two major categories: DA and DG. With the rapid advancement of deep learning, deep DA methods

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have become the mainstream choice for image classification tasks [8], [9], [10]. These methods typically design specific adaptation layers or strategies to reduce data distribution shifts between SD and TD, thereby enhancing the model performance on the TD [11], [12], [13]. For instance, Qu et al. [14] proposed an HSI classification method based on physically constrained shared abundance space. By projecting the data from SD and TD onto a shared abundance space according to their physical characteristics, this method significantly reduces spectral shifts between domains. In the context of cross-scene wetland mapping, Huang et al. [15] developed a spatial–spectral weighted adversarial DA (SSWADA) network, which significantly enhances wetland classification accuracy (CA). In addition, Li et al. [16] proposed a two-stage contrastive multilayer perceptron (MLP) network (MAC-CDZS) to address the zero-shot problem in the context of domain transfer. To further incorporate the prior information of spectral coordinates, Li et al. [17] proposed an asymmetric encoder–decoder architecture, named spectral coordinate Transformer (SCFormer). This model integrates spectral coordinate blocks equipped with random and sequential masking mechanisms to reduce interference from convolutional operations on spectral positioning. In addition, the Wasserstein distance (WD) is utilized to formulate an inter-domain loss function, facilitating effective domain alignment.

Compared with DA methods, research on DG in cross-domain HSI classification tasks is still relatively limited. A key distinction lies in the training data requirements: DA methods utilize labeled SD data along with unlabeled TD data, whereas DG is restricted to using only labeled SD data without any exposure to TD samples [18], [19], [20]. This constraint means DG methods cannot rely on TD priors to explicitly reduce spectral shifts, posing greater challenges for model generalization [21], [8], [22]. Despite these challenges, DG has seen notable advancements in computer vision [23], [24]. For example, Chen et al. [25] proposed rationale invariance DG (RIDG), which introduces an invariance loss to constrain rationale matrices of same-class samples during classification, thereby encouraging models to leverage domain-invariant features. Huang et al. [26] proposed the invariant directed acyclic graph (iDAG) search framework, which constructs class-conditional prototypes and designs a hierarchical contrastive learning module with clustering effects, contributing to enhanced prototypes and more stable predictions. Yu and Hwang [27] proposed attend to expert prompts (A2XP) to protect the privacy of network structures. In the first stage, prompts for each SD are optimized. In the second stage, two embedding networks are trained to effectively integrate expert prompts to obtain optimal outputs. In the context of cross-domain HSI classification, several DG methods have been recently proposed. Zhang et al. [28] proposed a single-SD expansion network (SDEnet) based on generative adversarial ideas. In this network, semantic and morphological encoders were designed in the generator to simulate TD data, and a discriminator was further designed based on contrastive learning to learn classwise domain-invariant representations. Zhang et al. [29] also introduced the language-aware DG network (LDGnet), incorporating text information as cross-domain shared prior knowledge. This method designed an image encoder and a text encoder to learn domain-invariant representations. To improve interaction between visual and linguistic modalities, Dang et al. [30] proposed the linguistic interact-with-visual engager (LIVE). In addition, Zhao et al. [31] presented the locally linear unbiased randomization network (LLURnet), which uses a symmetric encoder–decoder structure to implicitly construct local joint features under style randomization, effectively achieving domain expansion in the latent space [31].

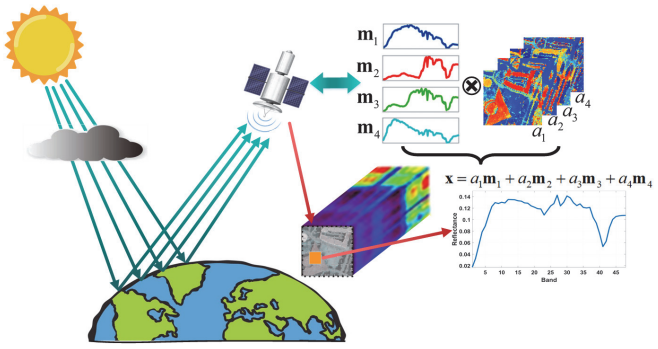


Fig. 1. Illustration of mixed pixels in HSI. A pixel is a linear combination of multiple endmembers in different proportions.

Hyperspectral unmixing aims to decompose mixed pixel spectra into a set of pure constituent spectra (endmembers) and their corresponding fractional abundances. With the rapid evolution of deep learning, autoencoder-based networks have emerged as powerful frameworks for nonlinear unmixing [32], [33]. To capture progressive spatial–spectral representations, multistage and cascaded architectures are developed. For instance, the multistage convolutional autoencoder network (MSNet) [34] utilizes a multistage convolutional autoencoder for hierarchical feature extraction, while the cycle-consistency unmixing network (CyCU-net) [35] introduces a cycle-consistency cascaded autoencoder to ensure spectral reconstruction fidelity. To further model complex spatial–spectral context, attention mechanisms and Vision Transformers have been widely integrated. Multimodal unmixing network (MUNet) [36] incorporates attention modules for multimodal unmixing, whereas abundance-guided attention spectral and spatial attention network (A2SAN) [37] employs abundance-guided spatial and spectral attention to refine abundance estimation. More recently, SRViT [38] leverages a self-supervised relation-aware Vision Transformer to capture global context and long-range spectral dependencies.

In the existing single-source DG methods, most of them are based on style transfer, photometric, and geometric transformations to generate out-of-domain samples, ignoring the spectral shift between scenes caused by the potential endmember variability of mixed pixels in HSI. This may lead to a lack of domain expansion strategies suitable for the multidimensionality of space-spectrum. Each pixel is usually a mixed pixel containing multiple substances in HSI. As shown in Fig. 1, each pixel is a linear combination of multiple endmembers in different proportions, where endmember variability can reflect spectral shifts between different scenes. Furthermore, utilizing shared feature extractors and alignment strategies to extract a shared space is currently the primary means by which DA and DG methods address domain shifts, as illustrated in

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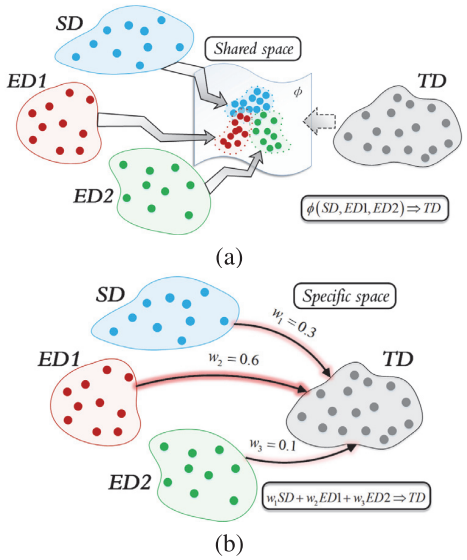


Fig. 2. Transfer learning strategies. (a) Extract the shared space and fit the TD using domain-invariant representations. (b) Extract the specific spaces and fit the TD using multidomain-specific representations.

Fig. 2(a), with the transfer paradigm depicted in Fig. 3(a). For example, DG methods such as RIDG, iDAG, and SDEnet generate out-of-domain samples and learn domain-invariant representations across multiple domains. However, existing methods ignore the difficulty in obtaining optimal solutions for the shared space in the presence of significant gaps between domains. Instead of focusing on extracting a shared space, we capture domain knowledge based on specific spaces and leverage collaborative multidomain-specific representations to infer TD, as shown in Fig. 2(b), with the transfer paradigm illustrated in Fig. 3(b).

In order to solve the above issues, a spectral unmixing-guided prompt network (SUPnet) is proposed for cross-domain HSI classification. Domain extension is carried out from the perspective of spectral unmixing, and multidomain-specific representation collaborative inference is realized by a prompt learning strategy. Specifically, SUPnet includes two components: a SUG and MPL. An unsupervised method is used to simultaneously extract the endmembers of SD and estimate abundances in the SUG. Furthermore, considering the close relationship between endmember variability and spectral shift, it simulates the variability of endmembers across different scenes by randomizing within the latent space and combines abundance information to generate ED data. Domain-level and endmember-level prompt learning is performed in MPL, where domain prompts and endmember prompts are designed to capture multidomain-specific representations (i.e., data knowledge and endmember knowledge). Furthermore, a domain adapter and an endmember adapter are designed to integrate the two types of prompts and sense how much domain-specific representation contributes to different domains. Finally, it is used to extract an appropriate prompt for TD inference. The main contributions of this article are as follows.

1) Considering the direct relationship between endmember variability and spectral shift between scenes. The SUG is designed in SUPnet to extract endmembers and abundances at the same time and to simulate endmember variability through randomization, achieving a spectral unmixing-guided domain expansion strategy.
2) To implement the transfer paradigm of multidomain-specific representations, the MPL strategy is developed. MPL includes prompt learning at two levels, introducing domain prompts and endmember prompts to capture data and endmember knowledge from different scenes.
3) For HSI data and its decomposed endmembers, a domain adapter and endmember adapter are designed to integrate them and update prompts to perceive the contribution of domain-specific representations to different domains.
4) We establish a well-coordinated mechanism for both unmixing and classification. Comprehensive experiments demonstrate that classification, as a downstream task, facilitates unsupervised spectral unmixing in large-scale scenes. Guided by spectral unmixing, SUPnet also exhibits superior performance compared with existing DA and DG methods.

The rest of this article is organized as follows. Section II elaborates on the proposed SUPnet. The extensive experiments and analyses on cross-domain HSI classification and unsupervised spectral unmixing are presented in Section III. Finally, conclusions are drawn in Section IV.

II. Spectral Unmixing-Guided Prompt Network

The main framework of our proposed SUPnet is composed of SUG (see Fig. 4) and MPL (see Fig. 5). The spatial patch sample in HSI is selected from the SD and sent to the SUG, where the endmembers of the SD are extracted and the abundance values are estimated. Variable endmembers are generated through randomization, and ED samples are synthesized using the abundance values as weights. Furthermore, the data and endmembers from both SD and ED are input into the Transformer-based MPL, where domain prompts and endmember prompts are used to learn multi-domain-specific representations. These prompts are updated through the corresponding adapters to assess the importance of domain-specific representations for inference TD.

A. Spectral Unmixing Generator

The linear mixing model (LMM) assumes that each pixel's spectrum is a linear combination of the spectra of several pure materials, known as endmembers. The coefficients of this linear combination must satisfy the abundance sum-to-one constraint (ASC) and the abundance nonnegativity constraint (ANC). In the LMM, the observed spectra $\mathbf{x}_i$ can be expressed as follows:

$$\mathbf{x}_i = \sum_{j=1}^{n_{em}} a_{ij}\mathbf{m}_j + \mathbf{e}_i$$
$$s.t. \sum_{j=1}^{n_{em}} a_{ij} = 1, a_{ij} \geq 0 \quad (1)$$

where $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{n}, \mathbf{x}_i \in \mathbb{R}^d$ represents the observed spectral vector for the ith pixel, where d is the dimension of spectral bands and n denotes the number of source samples. $\mathbf{a}_i = \{a_{ij}\}_{j=1}^{n_{em}}$ denotes the sequence of the abundance values corresponding to each endmember for the ith pixel, representing the proportion of each endmember in the observed

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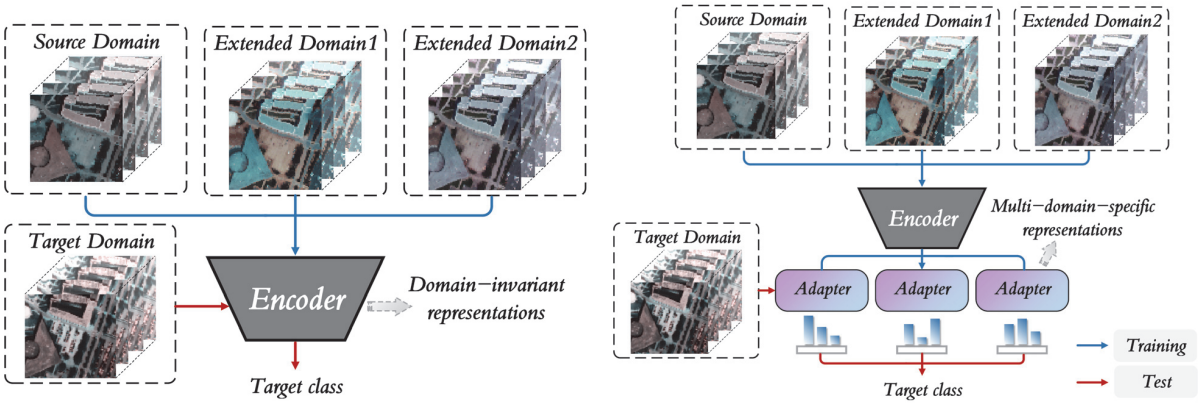


Fig. 3. Transfer paradigm of (a) domain-invariant representations and (b) multidomain-specific representations.

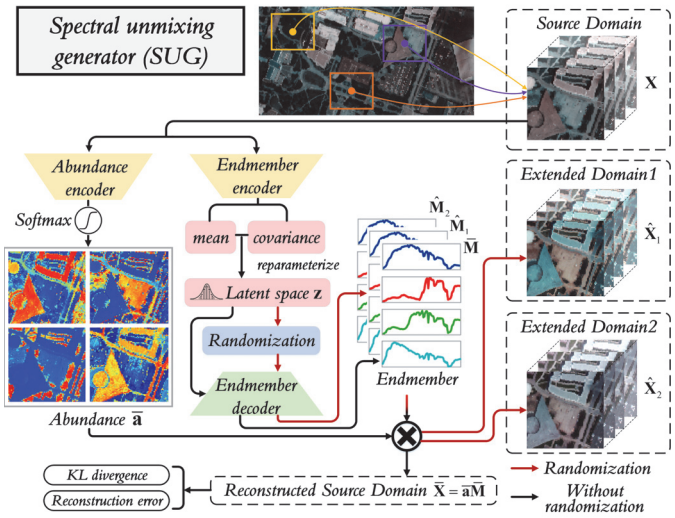


Fig. 4. Framework of SUG consists of an abundance encoder and an endmember encoder–decoder, which are composed of MLP–BN–ReLU. Within this framework, the red arrows represent operations with randomization. The latent space is randomized to generate variable endmembers, which are then linearly weighted with the abundances to obtain ED samples. In addition, the black arrows represent operations without randomization, extracting SD endmembers for calculating the reconstruction loss.

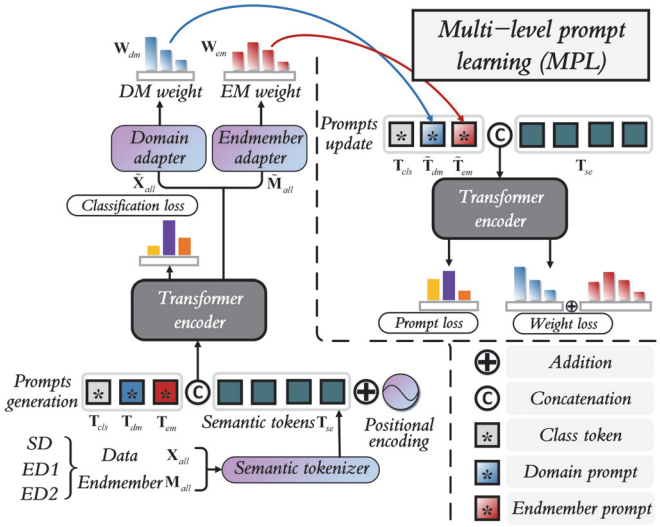


Fig. 5. Flowchart of MPL includes a semantic tokenizer, Transformer encoder, domain, and endmember adapters. The semantic tokenizer focuses on HSI properties and learns spatial–spectral projections to construct an HSI vocabulary set. Domain and endmember adapters are used to convert representations into weights for updating prompts.

spectrum. $\mathbf{M}_i = \{\mathbf{m}_j\}_{j=1}^{n_{em}}, \mathbf{m}_j \in \mathbb{R}^d$, is the endmember sequence of the ith pixel, $\mathbf{m}_j$ and $\mathbf{e}_i \in \mathbb{R}^d$ are the jth endmember and noise vectors for the ith pixel. n_{em} is the number of endmembers, which is set to the number of classes in the dataset. By satisfying the ASC and ANC constraints, the LMM provides a physically meaningful explanation that helps in decomposing and identifying the pure material components in HSI and quantifying their abundances.

1) Abundance Encoder: Combining the physical interpretability of the LMM, we design SUG to implement a domain extension strategy by simulating the variations in environmental conditions in TD through linear combinations of perturbed endmembers. The overall framework of SUG is illustrated in Fig. 4. Samples of $w \times h \times d$ (w and h are set to 13 in the experiment, i.e., $13 \times 13 \times d$) spatial patches in HSI are selected from SD and input separately into the abundance encoder, which estimates the abundance values, and the endmember encoder–decoder, which extracts and randomizes the endmembers. The abundance values are used as coefficients, and the perturbed endmembers are linearly combined to obtain multiple EDs. First, the abundance encoder performs a nonlinear transformation of the observed pixels

$$\bar{\mathbf{a}}_i = softmax\left(f_a\left(\mathbf{x}_i\right)\right) \tag{2}$$

where $\bar{\mathbf{a}}_i$ denotes the estimated abundance value, $f_a(\cdot)$ represents the abundance encoder consisting of MLP, BatchNorm, and ReLU (MLP–BN–ReLU), using a softmax layer to ensure the ASC and ANC constraints.

2) Endmember Encoder–Decoder: In the endmember encoder–decoder structure, we introduce a latent space that follows a standard normal distribution $\mathbf{z}_i \in \mathbb{R}^{d_z}$, d_z is the feature dimension of $\mathbf{z}_i$ (d_z is set to 64). The endmember generation process is simplified as follows:

$$\bar{\mathbf{M}}_i = f_z^{DC}\left(\mathbf{z}_i\right) \tag{3}$$

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where $\bar{\mathbf{M}} = \{\bar{\mathbf{M}}_i\}_{i=1}^{n}, \bar{\mathbf{M}}_i \in \mathbb{R}^{n_{em}\times d}$, represent the estimated endmembers, and $f_z^{DC}(\cdot)$ is a nonlinear function that associates the endmember with the underlying representation of the latent space, which is the endmember decoder composed of MLP–BN–ReLU in Fig. 4. The latent variable $\mathbf{z}_i$ can be interpreted as a compressed representation within the latent space of the SD endmembers. The observed pixel spectra are input into the endmember encoder to estimate the latent variables $\mathbf{z}_i$. However, the true posterior distribution $p(\mathbf{z}_i|\mathbf{x}_i)$ of $\mathbf{z}_i$ conditioned on pixel $\mathbf{x}_i$ is intractable. Variational inference is used to approximate and estimate $\mathbf{z}_i$. We treat the distribution of the endmember encoder $f_z^{ec}(\cdot)$ parameterized by θ as the variational distribution $q_\theta(\mathbf{z}_i|\mathbf{x}_i)$ for approximation of $p(\mathbf{z}_i|\mathbf{x}_i)$. To make the computation more feasible and efficient, $q_\theta(\mathbf{z}_i|\mathbf{x}_i)$ is constrained to be the probability density function (pdf) with a diagonal covariance matrix of Gaussian distribution $\mathcal{N}(\mu, \Sigma)$, where μ is the mean vector and $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_{d_z}^2)$. As shown in Fig. 4, each observed pixel is input into the endmember encoder (MLP–BN–ReLU) for a nonlinear transformation, representing the mean and the diagonal of the covariance matrix of the variational distribution, and the latent representation $\mathbf{z}_i$ of the endmembers is obtained through the reparameterization trick. To simplify the calculation of high-dimensional distributions, it is assumed that $\mathbf{z}_i$ is independent and identically distributed (i.i.d.) along both its components and index i. $p(\mathbf{z}_i)$ is constrained to a standard Gaussian pdf.

The goal of variational inference is to find the optimal variational parameters θ by minimizing the KL divergence between the true posterior distribution $p(\mathbf{z}_i|\mathbf{x}_i)$ and the variational distribution $q_\theta(\mathbf{z}_i|\mathbf{x}_i)$. The KL divergence is defined as follows:

$$KL\left(q_\theta\left(\mathbf{z}_i\middle|\mathbf{x}_i\right)\middle\|p\left(\mathbf{z}_i|\mathbf{x}_i\right)\right) = E_{q_\theta(\mathbf{z}_i|\mathbf{x}_i)}\left[\log\frac{q_\theta\left(\mathbf{z}_i\middle|\mathbf{x}_i\right)}{p\left(\mathbf{z}_i\middle|\mathbf{x}_i\right)}\right]. \tag{4}$$

To construct the evidence lower bound (ELBO)

$$\begin{aligned} &E_{q_\theta(\mathbf{z}_i|\mathbf{x}_i)}\left[\log p\left(\mathbf{x}_i\middle|\mathbf{z}_i\right) + \log p\left(\mathbf{z}_i\right) - \log q_\theta\left(\mathbf{z}_i\middle|\mathbf{x}_i\right)\right] \\ &= E_{q_\theta(\mathbf{z}_i|\mathbf{x}_i)}\left[\log p\left(\mathbf{x}_i|\mathbf{z}_i\right)\right] - KL\left(q_\theta\left(\mathbf{z}_i|\mathbf{x}_i\right)\|\,p\left(\mathbf{z}_i\right)\right). \end{aligned} \tag{5}$$

Since $E_{q_\theta(\mathbf{z}_i|\mathbf{x}_i)}[\log p(\mathbf{x}_i|\mathbf{z}_i)] \le 0$, maximizing the ELBO is approximately equivalent to minimizing $KL(q_\theta(\mathbf{z}_i|\mathbf{x}_i)\,\|\,p(\mathbf{z}_i))$. Here, the KL divergence can be expressed as the sum of KL divergences along each dimension of the latent space, and its closed-form solution is

$$\begin{aligned} \mathcal{L}_{KL} &= KL\left(q_\theta\left(\mathbf{z}_i\middle|\mathbf{x}_i\right)\middle\|p\left(\mathbf{z}_i\right)\right) \\ &= \sum_{j=1}^{d_z}\frac{1}{2}\left(\sigma_j^2 + \mu_j^2 - \log\sigma_j^2 - 1\right). \end{aligned} \tag{6}$$

As shown by the black arrows in Fig. 4, by inputting $\mathbf{z}_i$ into the endmember decoder, we estimate the endmembers for the observed pixel. In addition, we randomize $\mathbf{z}_i$ and input it into the endmember decoder to obtain the perturbed endmembers, as shown by the red arrows

$$\begin{aligned} \bar{\mathbf{M}}_i &= sigmoid\left(f_z^{DC}\left(f_z^{ec}\left(\mathbf{x}_i\right)\right)\right) \\ \bar{\mathbf{M}}_i &= sigmoid\left(f_z^{DC}\left(Random\left(f_z^{ec}\left(\mathbf{x}_i\right)\right)\right)\right) \end{aligned} \tag{7}$$

where $\hat{\mathbf{M}} = \{\hat{\mathbf{M}}_i\}_{i=1}^{2n}, \hat{\mathbf{M}}_i \in \mathbb{R}^{n_{em}\times d}$, represent the perturbed endmembers; there are two perturbations by default. In all experiments. $sigmoid(\cdot)$ is used to constrain the endmembers within (0, 1). $Random(\cdot)$ indicates the operation of random content replacement in a minibatch; its formula is as follows:

$$Random\left(\mathbf{z}_i, \mathbf{z'}_i\right) = dev\left(\mathbf{z}_i\right)\frac{\mathbf{z'}_i - mean\left(\mathbf{z'}_i\right)}{dev\left(\mathbf{z'}_i\right)} + mean\left(\mathbf{z}_i\right) \tag{8}$$

where $\mathbf{z'}_i$ is a randomly shuffled $\mathbf{z}_i$, $mean(\cdot)$ and $dev(\cdot)$ represent the mean and standard deviation, respectively. Furthermore, the ED data $\hat{\mathbf{X}}$ is obtained by weighting the perturbed endmembers $\bar{\mathbf{M}}_i$ with the abundance value, i.e., matrix vector multiplication

$$\hat{\mathbf{X}} = \sum_{i=1}^{2n}\bar{\mathbf{a}}_i\hat{\mathbf{M}}_i. \tag{9}$$

It is defaulted to obtain two ED data by random twice. After obtaining the reconstructed spectra by weighting the endmembers $\hat{\mathbf{M}}_i$ with the estimated abundance values, calculate the reconstruction loss between the reconstructed and observed pixel spectra

$$\mathcal{L}_{rec} = \sum_{i=1}^{n}\left\|\mathbf{x}_i - \bar{\mathbf{a}}_i\bar{\mathbf{M}}_i\right\|^2. \tag{10}$$

The KL divergence loss and reconstruction loss are used to optimize SUG and estimate accurate abundance values and endmembers

$$\mathcal{L}_{SUG} = \mathcal{L}_{KL} + \mathcal{L}_{rec}. \tag{11}$$

Finally, SUG outputs the data from the two EDs along with their corresponding perturbed endmembers, as well as the endmembers of SD. During the testing phase, the endmember of TD is directly estimated without perturbed operation.

B. Multilevel Prompt Learning

The primary objective of DG methods is to learn a robust model on one or multiple SDs that can be applied to TDs unseen during training. As shown in Fig. 3(a), the most common strategy is to learn domain-invariant representations through a shared feature extractor, embedding data from all domains into a common feature space. However, this approach ignores the difficulty that the model has in obtaining an optimal solution to the shared space at large spectral shifts. The transfer paradigm of multidomain-specific representation focuses on extracting the differences between various domains to achieve cooperative inference, as illustrated in Fig. 3(b). Inspired by this, the MPL strategy has been developed to learn more discriminative prompts within each domain and utilize domain-specific knowledge for more robust inference.

1) Semantic Tokenizer: The overall framework of MPL is illustrated in Fig. 5, where the data $\mathbf{X}$ and $\hat{\mathbf{X}}$ from SD and EDs, respectively, as well as the endmembers $\bar{\mathbf{M}}$ and $\hat{\mathbf{M}}$, are used as inputs. The semantic tokens for SD and EDs are obtained through a semantic tokenizer. Unlike ViT methods that generate tokens by dividing patches in the spatial dimension of the raw data, this semantic tokenizer utilizes learned spatial–spectral projections to encode the spatial–spectral information to generate tokens. First, a spatial–spectral extractor obtains the projection matrix through a series of Conv3d layers and Conv2d layers. Subsequently, a softmax function is applied to the projection matrix to compute the spatial–spectral attention maps. A Conv2d layer with 1×1 convolutional kernels is incorporated to finalize the dimension

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mapping, yielding compact vocabulary sets of size L (L is set to 6 in all experiments), referred to as semantic tokens $\mathbf{T}_{se} \in \mathbb{R}^{L\times d_{se}}$ (d_{se} is set to 64). Formally

$$\mathbf{T}_{se} = softmax(f_{se}(\mathbf{X}_{all}; \mathbf{W}_{se}))^T \mathbf{X}_{all} \tag{12}$$

where $\mathbf{X}_{all} \in \mathbb{R}^{(n_{dm}\times n)\times d}$ represents the concatenation of SD data and ED data $[\mathbf{X}; \hat{\mathbf{X}}]$. n_{dm} denotes the number of domains used for training, which is set to 3 in the experiment (including SD, ED1, and ED2). $f_{se}(\cdot)$ refers to the spatial–spectral extractor, which is composed of Conv3d and Conv2d layers with learnable kernels $\mathbf{W}_{se}$.

2) Prompt Generation: Domain-level and endmember-level prompts are assigned to each domain with prompt generation. We construct labels $\mathbf{Y}_{dm} \in \mathbb{R}^{n_{dm}\times n}$ and $\mathbf{Y}_{em} \in \mathbb{R}^{n_{dm}\times n\times n_{em}}$, respectively, for the data and endmembers of SD and ED according to their domain categories, assigning the same label to prompts within the same domain (the label value ranges from 0 to 2). Here, the labels $\mathbf{Y}_{dm}$ and $\mathbf{Y}_{em}$ are the domain indices, not semantic class labels. These labels are then parameterized to obtain domain prompts $\mathbf{T}_{dm} \in \mathbb{R}^{L\times d_{se}}$ and endmember prompts $\mathbf{T}_{em} \in \mathbb{R}^{L\times d_{se}}$

$$\begin{aligned} \mathbf{T}_{dm} &= W_1(\mathbf{Y}_{dm}) \\ \mathbf{T}_{em} &= mlp(W_2(\mathbf{Y}_{em})) \end{aligned} \tag{13}$$

where $W_1(\cdot)$ and $W_2(\cdot)$ are the learnable embedding layers used to map discrete domain indices into continuous and dense vector representations. $mlp(\cdot)$ is used to project the parameterized label $W_2(\mathbf{Y}_{em})$. The two levels of learnable prompts are used to capture domain-specific knowledge, preventing the model from focusing solely on extracting the domain-invariant feature. All tokens are combined as $[\mathbf{T}_{cls}; \mathbf{T}_{dm}; \mathbf{T}_{em}; \mathbf{T}_{se}]$ and input into the Transformer encoder along with learnable positional embeddings (PEs), which includes a learnable classification token $\mathbf{T}_{cls} \in \mathbb{R}^{d_{se}}$. Through the attention mechanism, all tokens are integrated, and the output class token representations are used to compute the classification loss

$$\mathcal{L}_{cls}(\mathbf{X}_{all}, \mathbf{Y}_{all}) = \frac{1}{n_{dm}\times n}\sum_{i=1}^{n_{dm}\times n} \mathcal{L}_{ce}(f_T(f_{se}(\mathbf{x}_i)), \mathbf{y}_i) \tag{14}$$

where $\mathcal{L}_{ce}$ denotes the cross-entropy loss, $f_T(\cdot)$ is the Transformer encoder, $\mathbf{Y}_{all} \in \mathbb{R}^{n_{dm}\times n}$ represents the concatenation of SD and EDs class labels. After the Transformer encoder is optimized by the above classification loss, the data and endmember of SD and EDs are input to obtain the corresponding representations $\tilde{\mathbf{X}}_{all} = f_T(f_{se}([\mathbf{X}; \hat{\mathbf{X}}]))$ and $\tilde{\mathbf{M}}_{all} = f_T(f_{se}([\tilde{\mathbf{M}}; \hat{\mathbf{M}}]))$.

3) Prompt Learning: A domain adapter and an endmember adapter are designed in SUPnet to perceive the relationship between multilevel domain-specific knowledge of SD and TD, and learn suitable adapted prompts for TD inference. Since TD is not involved in the training phase, we treat the training set as out-of-distribution data to simulate the TD inference process. First, the DM weights $\mathbf{W}_{dm} = \{\mathbf{w}_{dm}^i\}_{i=1}^{(n_{dm}\times n)}, \mathbf{w}_{dm}^i \in \mathbb{R}^L$, and EM weights $\mathbf{W}_{em} = \{\mathbf{w}_{em}^i\}_{i=1}^{(n_{dm}\times n\times n_{em})}, \mathbf{w}_{em}^i \in \mathbb{R}^L$, are obtained through the two adapters and used to update the prompts

$$\tilde{\mathbf{T}}_{dm} = A_{dm}(\tilde{\mathbf{X}}_{all})\mathbf{T}_{dm} = \sum_{i=1}^{n_{dm}\times n} \mathbf{w}_{dm}^i \mathbf{T}_{dm}$$

$$\tilde{\mathbf{T}}_{em} = A_{em}(\tilde{\mathbf{M}}_{all})\mathbf{T}_{em} = \sum_{i=1}^{n_{dm}\times n\times n_{em}} \mathbf{w}_{em}^i \mathbf{T}_{em} \tag{15}$$

where $A_{dm}(\cdot)$ and $A_{em}(\cdot)$ represent the domain adapter and endmember adapter, respectively. They are composed of MLPs with softmax, ensuring that the DM weights and EM weights satisfy nonnegative condition. These two adapters learn adaptive weights by analyzing the relationships between multiple domains and between endmembers, assigning larger weights to the prompts that are more relevant to the out-of-distribution data. Subsequently, the adapted prompts $\tilde{\mathbf{T}}_{dm}$ and $\tilde{\mathbf{T}}_{em}$ are combined with learnable classification tokens $\mathbf{T}_{cls}$ and $\mathbf{T}_{se}$ in (12) and input into the Transformer encoder to complete the inference on out-of-distribution data. To achieve effective prompt learning, we design the following two losses. On the one hand, when inferring out-of-distribution data, any domain in the training set may contribute to it. Therefore, the domain adapter and endmember adapter should capture adapted prompts that are helpful for the final prediction. The prompt loss is calculated as follows:

$$\mathcal{L}_{prompt}(\tilde{\mathbf{X}}_{all}, \mathbf{Y}_{all}) = \frac{1}{n_{dm}\times n}\sum_{i=1}^{n_{dm}\times n} \mathcal{L}_{ce}(f_T(f_{se}(\tilde{\mathbf{x}}_i)), \mathbf{y}_i). \tag{16}$$

On the other hand, the prompts that are strongly related to out-of-distribution data should be specific prompts during the training process, that is, the domain prompts and endmember prompts corresponding to their in-domain and out-of-domain counterparts of the same class. Therefore, the binary cross entropy of DM weights and EM weights are calculated separately to form the weight loss

$$\mathcal{L}_{weight} = \mathcal{L}_{bce}(\mathbf{W}_{dm}, \mathbf{B}_{dm}) + \mathcal{L}_{bce}(\mathbf{W}_{em}, \mathbf{B}_{em}) \tag{17}$$

where $\mathcal{L}_{bce}$ represents the binary cross-entropy loss, and $\mathbf{B}_{dm}$ and $\mathbf{B}_{em}$ are the binary labels for the domain prompt and endmember prompt. The value of a binary label is either 1 or 0, where 1 (0) indicates the positive (negative) class, meaning the weight is associated (not associated) with information from in-domain and out-of-domain samples of the same (different) class.

During the testing phase, the representations of TD data and endmembers are extracted and input into the fully trained adapter-aware Transformer encoder to perceive the correlation between TD and SD (or EDs). The domain prompt and endmember prompt of TD are then represented as linear combinations of different domains in the training set. These prompts, along with the classification token and semantic tokens, are input into the Transformer encoder to obtain the prediction results for TD.

Integrating the above loss functions, the total loss of SUPnet is defined as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda_1(\mathcal{L}_{rec} + \mathcal{L}_{KL}) + \lambda_2(\mathcal{L}_{prompt} + \mathcal{L}_{weight}) \tag{18}$$

where λ_1 and λ_2 are the hyperparameters for balancing the SUG and MPL.

III. Experimental Results and Analysis

In this section, the MFF, Houston and Paiva airborne dataset, Natural willow UAV dataset, and GID satellite dataset are utilized to validate the effectiveness of the proposed SUPnet

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TABLE I

NUMBER OF SOURCE AND TARGET SAMPLES FOR THE MFF AIRBORNE DATASET

Dataset	Class	Source	Target1	Target2	Total
		MFF-SD	MFF-TD1	MFF-TD2	
MFF	Larch	7,214	10,919	25,691	43,824
	Mongolian pine	4,023	9,773	15,782	29,578
	Korean pine	7,072	8,179	14,758	30,009
	Spruce	1,201	10,501	9,072	20,774
	Broad-leaved trees	1,753	10,311	12,636	24,700
	Total	21,263	49,683	77,939	148,885

model. To benchmark its performance, we incorporate several classic and state-of-the-art Transformer-based methods and cross-domain classification algorithms, including group-aware hierarchical Transformer (GAHT) [39], spatial-aware conformal prediction (SACP) [40], confident-learning-based DA (CLDA) [41], supervised contrastive-learning-based unsupervised DA (SCLUDA) [42], consistency-aware customized learning (CACL) [22], shift-sensitive spatial–spectral disentangling learning (S4DL) [43], Fourier-based spectral–spatial generator (FSSG) [44], SDEnet [28], and A2XP [27]. CA, overall accuracy (OA), average accuracy (AA), and Kappa Coefficient (KC) are employed to evaluate the classification performance.

A. Experimental Data

We adopt five cross-domain datasets in experiments, including the MFF cross-temporal airborne dataset, Houston cross-temporal airborne dataset, GID cross-scene satellite dataset, Pavia cross-scene airborne dataset, and Natural Willow cross-temporal cross-region UAV dataset. The land cover classes and the number of samples for all datasets are detailed in Tables I and II.

B. Ablation Studies

1) Influence of λ_1 and λ_2: In the context of the outlined SUPnet framework, the adjustable parameters are the regularization parameters, specifically λ_1 and λ_2 in (18), which are strategically chosen from a predefined set comprising $1e^{-3}$, $1e^{-2}$, $1e^{-1}$, 1, $1e^{+1}$. These λ_1 and λ_2 serve as critical hyperparameters, modulating the relative influence of SUG and MPL within the SUPnet architecture. Fig. 6 visually illustrates the trend in the OA of the SUPnet model across the four experimental datasets with various combinations of λ_1 and λ_2. This visualization highlights the sensitivity of the specific selection of these hyperparameters to the OA. The optimal values of parameters λ_1 and λ_2 for the MFF-TD1, MFF-TD2 are determined to be 1 and $1e^{-1}$, Houston 2018, GID-wh and PU are $1e^{-1}$ and 1.

2) Influence of n_{dm}: We analyze the impact of the number of EDs generated by SUG on the proposed method. n_{dm} denotes the number of domains used for training, which includes one SD and multiple EDs. The selection range for n_{dm} is 1–5. When $n_{dm} = 1$, SUG does not generate ED, and only the abundance values and endmembers of the SD are estimated and input into MPL. Table III presents the classification accuracies corresponding to different values of n_{dm}. When $n_{dm} = 3$, which means two perturbations are applied to the endmembers in SUG to obtain two EDs, SUPnet achieves the best performance on the three target scenes (MFF-TD1, MFF-TD2, and Houston 2018). On GID-wh, the best performance is observed when $n_{dm} = 4$, with $n_{dm} = 3$ yielding an accuracy that is 0.49% lower. To maintain a consistent setting, n_{dm} is fixed at 3 for all experiments.

TABLE II

NUMBER OF SOURCE AND TARGET SAMPLES FOR HOUSTON, GID, PAVIA, AND NATURAL WILLOW DATASETS

Dataset	Class	Source	Target	Total
Houston	-	Houston 2013	Houston 2018	-
	Grass healthy	345	1,353	1,698
	Grass stressed	365	4,888	5,253
	Trees	365	2,766	3,131
	Water	285	22	307
	Residential buildings	319	5,347	5,666
	Non-residential buildings	408	32,459	32,867
	Road	443	6,365	6,808
	Total	2,530	53,200	55,730
GID	-	GID-nc	GID-wh	-
	Rural residential	5,495	4,729	10,224
	Irrigate land	3,643	5,643	9,286
	Garden Land	6,171	6,216	12,387
	River	2,858	11,558	14,416
	Lake	5,172	2,666	7,838
	Total	23,339	30,812	54,151
Pavia	-	PC	PU	-
	Tree	7,598	3,064	10,662
	Asphalt	9,248	6,631	15,879
	Brick	2,685	3,682	6,367
	Bitumen	7,287	1,330	8,617
	Shadow	2,863	947	3,810
	Meadow	3,090	18,649	21,739
	Bare soil	6,584	5,029	11,613
	Total	39,355	39,332	78,687
Natural Willow	-	NW-AreaI	NW-AreaII	-
	Natural willow	8,808	15,116	23,924
	Tamarix chinensis	2,909	13,301	16,210
	Yellow River water	4,003	7,306	11,309
	Highway	1,114	1,381	2,495
	Earth road	4,982	486	5,468
	Ecological restoration pool	9,084	11,212	20,296
	Reed	32,896	22,143	55,039
	Total	63,796	70,945	134,741

TABLE III

PARAMETER TUNING OF n_{dm} FOR THE PROPOSED SUPNET USING ALL THE FOUR EXPERIMENTAL DATA

Target scene OA (%)	n_{dm}				
	1	2	3	4	5
MFF-TD1	68.30	69.88	**71.12**	70.87	69.04
MFF-TD2	70.72	73.85	**75.16**	74.69	71.91
Houston 2018	78.16	78.44	**81.54**	79.07	80.08
GID-wh	79.75	81.89	85.22	**85.71**	83.34

3) Influence of SUG and MPL: SUPnet is mainly composed of two key modules, SUG and MPL. An ablation study is conducted by removing each module independently, as shown in Table IV. To ensure a fair comparison, model parameters are adjusted via the semantic token dimension d_{se} and the depth of the Transformer encoder blocks. Retaining only the Transformer encoder and classification loss (baseline without SUG or MPL) yields 65%~67% accuracy on four datasets (No. 1). Adding SUG enhances performance by 3%~6% via spectral-unmixing-guided domain expansion (No. 3). Incorporating MPL without SUG (using offline-extracted endmembers) improves accuracy by 4.4%~7.6% (No. 5). The full SUPnet model with both SUG and MPL achieves the best performance (No. 7). Parameter scaling (e.g., No. 1 versus 2) yields only minor gains, indicating that performance is driven by architecture

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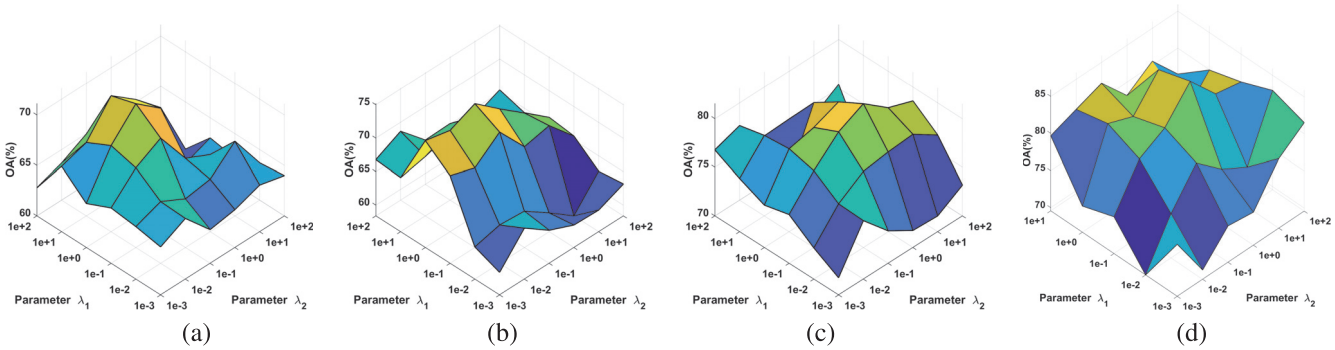


Fig. 6. Parameter tuning of λ_1 and λ_2 for the proposed SUPnet using all four experimental data. (a) MFF-TD1. (b) MFF-TD2. (c) Houston 2018. (d) GID-wh.

TABLE IV
ABLATION COMPARISON OF EACH MODULE OF SUPNET

Ablation Setting			OA (%)				Params(M)
No.	SUG	MPL	MFF-TD1	MFF-TD2	Houston 2018	GID-wh	
1	×	×	66.02	67.44	65.76	74.98	0.64
2	×	×	66.95	68.13	66.81	75.72	3.55
3	√	×	69.51	70.62	71.73	78.05	1.18
4	√	×	69.88	70.03	71.22	78.26	3.03
5	×	√	70.43	72.10	73.38	81.43	2.63
6	×	√	70.48	72.63	73.02	81.38	3.25
7	√	√	**71.12**	**75.16**	**81.54**	**85.22**	3.17

TABLE V
ABLATION COMPARISON OF DOMAIN ADAPTER AND ENDMEMBER ADAPTER IN MPL

Ablation Setting			OA (%)				Params(M)
No.	Domain adapter	Endmember adapter	MFF-TD1	MFF-TD2	Houston 2018	GID-wh	
1	×	×	69.88	70.03	71.22	78.26	3.03
2	√	×	70.92	74.44	79.37	84.03	1.26
3	×	√	70.65	73.28	77.86	82.61	3.09
4	√	√	**71.12**	**75.16**	**81.54**	**85.22**	3.17

rather than parameter count. Under comparable parameters (~3.17M), the full model (75.16% OA on Houston 2018) significantly outperforms all ablated variants (68.13%, 70.03%, and 72.63%), demonstrating the critical and complementary roles of SUG and MPL.

4) Influence of the Domain Adapter and Endmember Adapter in MPL: We analyze the roles of the domain adapter and endmember adapter in MPL, with the ablation experiments presented in Table V. All variants in the ablation experiments include SUG. No. 1 only contains SUG and a Transformer encoder. No. 2 incorporates prompt learning with only the domain-level prompt and its corresponding adapter. Similarly, No. 3 integrates prompt learning with only the endmember-level prompt and its associated adapter. No. 4 refers to the SUPnet equipped with MPL. Compared with the baseline (which uses no adapters at all), the standalone application of either the domain adapter or the endmember adapter leads to a significant improvement in OA across multiple datasets. Among these, the use of only the domain-level prompt and adapter consistently outperforms that of the endmember-level prompt and adapter across all datasets. This phenomenon arises because the intrinsic information of the data plays a dominant role in generalized learning. When the two adapters are used in combination, the model achieves the best performance on all datasets. This indicates that the domain adapter and endmember adapter exhibit a complementary effect, enabling them to enhance the discriminative capability of multidomain-specific representations from different levels.

TABLE VI
ABLATION COMPARISON OF PROMPT LEARNING IN MPL

Ablation Setting	MFF-TD1	MFF-TD2	Houston 2018	GID-wh	Params(M)
Baseline	66.95	68.13	66.81	75.72	3.55
Random Prompts	68.10	67.96	70.04	76.85	3.17
MPL	**70.48**	**72.63**	**73.02**	**81.38**	3.25

5) Influence of Prompt Learning in MPL: To clarify the contribution of the prompt learning mechanism within the MPL module, we conduct an ablation study as shown in Table VI. In the Random Prompts setting, while maintaining the same Transformer encoder and input data as in the full MPL, the domain and endmember prompts are replaced with randomly initialized. In addition, the domain/endmember adapters and their associated loss functions ($\mathcal{L}_{prompt}$, $\mathcal{L}_{weight}$) are removed. The results clearly demonstrate that while the Random Prompts setting shows a slight improvement over the Baseline, this is attributable to the increased data diversity provided by the SUG. However, the full MPL model significantly outperforms it under identical architectural and input conditions (e.g., 72.63% versus 67.96% on MFF-TD2). This indicates that the performance gain stems not merely from the expressiveness of the Transformer or data augmentation, but is primarily due to the proposed prompt learning mechanism itself, which dynamically perceives and fuses multidomain-specific representations, thereby effectively enhancing generalization capability.

Influence of the unmixing network. To investigate the integration method and influence of the unmixing network, we conducted an ablation study comparing two-stage versus one-stage (end-to-end) integration, with results summarized in Table VII. The ablation experimental setup is as follows.

1) *Two Stages:* We first trained an unsupervised spectral unmixing network (A2SAN, MSNet, and the proposed SUG) separately using only the SD data. For A2SAN and MSNet, which lack an inherent domain expansion mechanism, we generated EDs by adding noise via adaptive instance normalization (AdaIN). SUG generated EDs through endmember perturbation. Then, the estimated abundances, endmembers, and generated ED

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TABLE VII

ABLATION COMPARISON OF THE TWO-STAGE TRAINING AND ONE-STAGE TRAINING FOR THE UNMIXING NETWORK

Ablation Setting		OA (%)				Params(M)
		MFF-TD1	MFF-TD2	Houston 2018	GID-wh	
Baseline		66.95	68.13	66.81	75.72	3.55
two-stage	A2SAN	69.64	72.11	73.47	80.20	3.25
	MSNet	70.15	72.64	72.29	80.87	3.25
	SUG	70.48	72.63	73.02	81.38	3.25
one-stage	SUPnet	**71.12**	**75.16**	**81.54**	**85.22**	3.17

data from each model were fed into the MPL module for training.

2) *One Stage:* This represents the full SUPnet, where the SUG and MPL modules are trained end-to-end in a coordinated manner.

Compared with the baseline, all two-stage methods have improved performance, which confirms that the expansion strategy based on abundance and endmembers is beneficial to the generalization of the model to the TD. Notably, in this two-stage setting, SUG generally performs on par with or slightly better than other two unmixing networks, A2SAN and MSNet, demonstrating its competitive unmixing capability. Furthermore, across all datasets, the single-stage SUPnet significantly and consistently outperforms all two-stage variants. This indicates that the final classification results have been greatly improved, which is not only due to the enhancement of unmixing quality but also largely attributed to the end-to-end collaboration mechanism between unmixing and classification.

C. Performance on Cross-Domain HSI Classification

1) Evaluation Setup: To assess the performance of SUPnet using only SD for training, we compare it with several algorithms, including GAHT, SACP, CLDA, SCLUDA, CACL, S4DL, FSSG, SDEnet, and A2XP. The setup for training samples is as follows: for DA methods like CLDA, SCLUDA, CACL, and S4DL, 180 labeled SD data (the rest are used for validation) along with some unlabeled TD data are used for training. In the case of GAHT, SACP, FSSG, SDEnet, A2XP, and SUPnet, only labeled SD data are used (also 180 labeled SD data are used for training, and the rest are used for validation). Table VIII reports the results across five datasets.

2) Quantitative Comparison: The results and corresponding analysis are discussed as follows.

1) The best performance among all comparison methods is achieved by CLDA with 68.26% on MFF TD1, 70.19% on MFF TD2, and 84.19% on GID-wh, and SDEnet reached 79.96% on Houston 2018, 72.57% on PU, and 87.06% on NW-AreaII. It is evident that unsupervised deep DA and DG methods have a significant advantage in cross-domain HSI classification tasks compared with methods like SACP and GAHT, which lack feature or distribution alignment strategies.
2) SUPnet demonstrates significant advantages compared with other DA and DG methods, especially on the MFF airborne dataset (a 4.97% increase in MFF TD2) and Natural willow UAV dataset (a 5.9% increase in NW-AreaII). DA methods such as CLDA, SCLUDA, and S4DL incorporate various pseudo-label strategies to select high-quality TD data for training, and heavily rely on feature alignment to mitigate spectral shift. DG methods, including RIDG, iDAG, and SDEnet, focus on generating out-of-domain samples and learning domain-invariant representations across multiple domains. In SUPnet, the spectral unmixing that aligns better with the characteristics of HSI data is employed to expand the diversity of spectral shifts. Instead of enforcing alignment on features with large distribution differences, we design the MPL module to extract multidomain-specific representations for more effective inference.
3) In the fine-grained recognition scenario of the MFF airborne dataset, SUPnet improves over CLDA by 2.86% and 4.97% on MMF TD1 and MMF TD2, respectively. The main idea of CLDA involves designing a confident-learning-based data analysis strategy to generate pseudo-labels for TD samples and selecting high-confidence samples. However, the strategy of estimating pseudo-labels for TD often fails due to potential spectral shifts between SD and TD. Instead of accessing TD, SUPnet simulates endmember variability to extend SD so that the model is fully trained under a variety of spectral shift conditions.
4) In urban scenes, SUPnet achieved improvements of 1.58%, 5.79%, and 1.03%, respectively, compared with the best-performing SDEnet (Houston 2018 and PU) and CLDA (GID-wh). This performance gain demonstrates the advantage of SUPnet's MPL mechanism over SDEnet's shared encoder approach. While SDEnet extracts domain-invariant features, SUPnet effectively mines domain-specific representations through dual-level prompt learning, enabling collaborative multidomain inference.
5) In the coastal wetland scene of the Natural willow UAV dataset, SUPnet achieves an OA of 92.96%, significantly outperforming the best baseline (SDEnet,

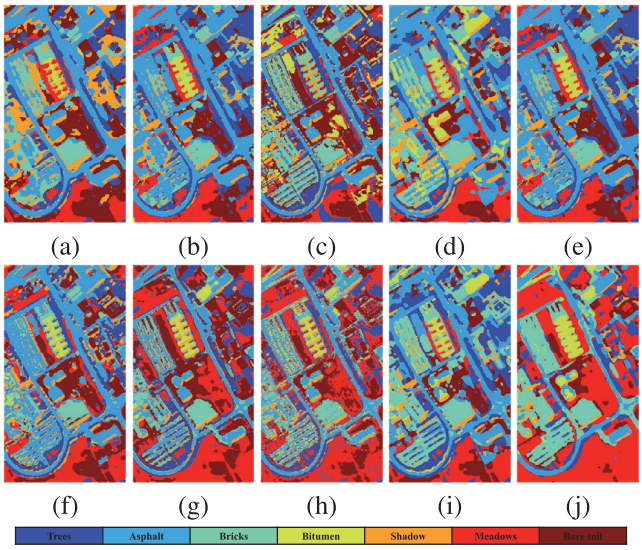


Fig. 7. Data visualization and classification maps for the target scene Pavia University data obtained with different methods. (a) GAHT (57.41%). (b) SACP (62.89%). (c) CLDA (60.67%). (d) SCLUDA (57.35%). (e) CACL (68.51%). (f) S4DL (64.79%). (g) FSSG (71.45%). (h) SDEnet (72.57%). (i) A2XP (61.87%). (j) SUPnet (78.36%).

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TABLE VIII
CLASS-SPECIFIC AND OVERALL CA (%) OF DIFFERENT METHODS FOR ALL TARGET SCENE DATA

Dataset	Class	Classification algorithms									
		GAHT [40]	SACP [41]	CLDA [42]	SCLUDA [43]	CACL [23]	S4DL [44]	FSSG [45]	SDEnet [29]	A2XP [28]	SUPnet
MFF-TD1	1	84.88	76.87	76.58	76.31	79.22	84.17	93.27	86.36	78.82	**95.57**
	2	82.67	83.24	92.61	**93.62**	83.17	82.18	77.68	88.96	76.84	84.59
	3	68.38	67.01	51.91	54.46	70.08	58.19	64.20	**78.10**	43.21	32.39
	4	43.95	38.77	42.65	13.85	39.75	28.21	34.04	33.39	47.12	**60.38**
	5	57.46	52.89	75.41	75.40	55.20	**82.38**	69.67	37.73	69.24	73.07
	OA (%)	67.38	63.18	68.26	62.72	66.86	67.30	68.00	64.23	63.88	**71.12**
	Kappa (κ)	59.22	54.24	60.31	53.55	58.62	59.14	60.04	55.40	54.80	**63.48**
MFF-TD2	1	73.05	69.71	76.41	82.32	82.49	78.59	**91.31**	72.64	76.88	89.75
	2	82.51	78.13	80.93	73.53	71.03	64.97	65.54	**82.61**	65.68	71.45
	3	80.65	75.77	77.03	84.08	76.45	81.85	73.49	78.86	84.12	**84.65**
	4	23.23	16.55	**26.66**	8.12	20.12	10.98	3.94	18.38	4.24	19.71
	5	60.72	61.75	67.42	40.84	54.87	**86.44**	63.75	65.23	69.57	78.86
	OA (%)	68.61	65.38	70.19	66.21	68.49	69.85	68.77	68.32	66.82	**75.16**
	Kappa (κ)	59.26	53.74	61.27	54.66	59.23	60.74	59.43	58.90	55.25	**67.27**
Houston 2018	1	23.21	73.37	68.07	**93.94**	42.34	12.07	73.39	24.69	47.43	72.14
	2	69.74	66.35	85.45	54.34	81.65	**89.59**	51.88	84.98	69.15	73.02
	3	**67.61**	58.74	60.85	57.85	46.01	56.83	48.59	59.65	58.35	49.39
	4	81.82	88.71	72.73	86.67	65.41	**100**	**100**	**100**	**100**	**100**
	5	92.37	80.01	93.57	75.87	71.05	45.80	**93.75**	62.33	52.91	88.65
	6	76.50	76.53	69.72	87.30	90.47	83.00	84.53	90.54	85.56	**92.20**
	7	47.18	58.38	**77.75**	60.02	45.37	59.03	46.61	57.45	52.34	43.68
	OA (%)	72.15	74.73	74.02	78.61	79.10	77.39	75.78	79.96	78.90	**81.54**
	Kappa (κ)	57.11	61.92	61.54	64.63	65.03	63.19	62.97	65.15	64.82	**68.66**
GID-wh	1	79.42	60.47	70.80	86.17	59.93	63.04	48.30	**91.06**	31.40	89.93
	2	**88.61**	75.20	73.33	59.29	78.70	82.47	60.84	35.18	75.21	82.76
	3	39.08	43.18	71.67	65.77	65.35	53.78	64.16	78.39	**98.63**	54.36
	4	99.62	99.47	98.08	99.21	99.38	**99.77**	99.63	95.10	97.08	99.58
	5	28.51	80.80	**99.89**	55.55	90.32	81.43	99.44	88.41	97.30	91.71
	OA (%)	76.14	76.82	84.19	79.38	81.89	80.10	77.47	77.95	83.32	**85.22**
	Kappa (κ)	67.51	68.11	78.97	74.20	75.84	74.19	72.37	72.91	77.76	**80.14**
Pavia University	1	83.75	85.33	**97.68**	82.31	84.07	93.05	89.30	91.12	96.02	65.57
	2	99.86	99.61	98.85	79.31	**100**	99.98	95.14	78.89	95.96	87.22
	3	56.41	41.20	**95.46**	66.87	22.51	19.45	43.97	55.24	93.16	90.75
	4	2.18	25.12	10.14	**96.99**	20.57	0.38	37.97	58.20	34.51	62.56
	5	94.51	94.51	**100**	85.32	74.76	94.93	96.09	99.37	96.94	73.92
	6	34.20	50.54	30.67	42.22	62.71	56.78	69.92	75.64	39.55	**85.29**
	7	**80.41**	64.98	58.72	46.59	73.99	75.46	52.77	53.01	56.60	44.75
	OA (%)	57.41	62.89	60.67	57.35	68.51	64.79	71.45	72.57	61.87	**78.36**
	Kappa (κ)	47.99	53.74	51.16	47.18	57.16	54.25	60.21	62.48	52.99	**69.09**
NW-AreaII	1	92.19	45.02	87.44	88.79	98.44	**98.18**	70.93	84.15	92.44	87.21
	2	2.38	13.08	11.26	80.03	11.39	7.19	70.86	54.60	23.91	**82.83**
	3	94.14	99.51	93.47	**100**	90.43	94.76	**100**	98.62	**100**	**100**
	4	4.42	73.71	**100**	**100**	97.25	98.33	99.93	79.15	86.97	99.93
	5	10.44	84.16	98.97	67.28	**100**	**100**	88.68	**100**	**100**	**100**
	6	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
	7	98.39	99.35	98.19	51.92	93.24	**99.86**	88.62	98.41	91.96	96.49
	OA (%)	76.88	70.54	79.44	78.64	79.91	81.60	84.71	87.06	81.36	**92.96**
	Kappa (κ)	69.79	63.37	73.12	73.28	73.91	75.99	80.51	83.34	75.91	**91.03**

87.06%). Notably, SUPnet reaches 82.83% accuracy in classifying spectrally similar "Tamarix chinensis" and "Natural Willow" categories, proving its enhanced fine-grained discrimination through endmember perturbation and prompt learning.

3) Qualitative Comparison: For further visual comparison, the classification maps for PU and NW-AreaII are presented in Figs. 7 and 8. In the classification maps of NW-AreaII, it can be observed that most comparison methods have serious confusion in distinguishing the first class (Natural willow), the second class (Tamarix chinensis), and the seventh class (Reed). Taking the method shown in Fig. 8(f) as an example, it can only identify part of the first class features, and at the same time, there is a significant misjudgment between the second class and the seventh class, and it almost fails to effectively identify the second class vegetation areas. In contrast, as can be seen from Fig. 8(j), the SUPnet can clearly distinguish the above three types of features, with more accurate recognition results, and the extracted spatial distribution and boundaries of wetland vegetation are more reasonable, showing better interclass discrimination ability and scene adaptability.

D. Effectiveness of Spectral Unmixing-Guided Domain Expansion

To verify the reliability of the proposed domain expansion strategy, we indirectly reflect it by demonstrating the relationship between endmember variability and spectral shift.

1) Quantitative Comparison: We calculate multiple spectral angle distance (SAD) values for Houston and GID datasets, including SAD between SD and ED endmembers (EM-SD, EM-ED), SAD between SD/ED samples (SD, ED), and SAD between SD/TD samples (SD, TD). All endmembers are extracted via the fully trained SUPnet, and SAD computations (both for endmembers and samples) are done categorywise. The results in Tables IX and X demonstrate that while endmember differences between SD and ED are minimal, significant spectral shifts exist between their samples. Comparative analysis of SAD values for (SD, ED) and (SD, TD) pairs reveals that the ED exhibits spectral deviations from

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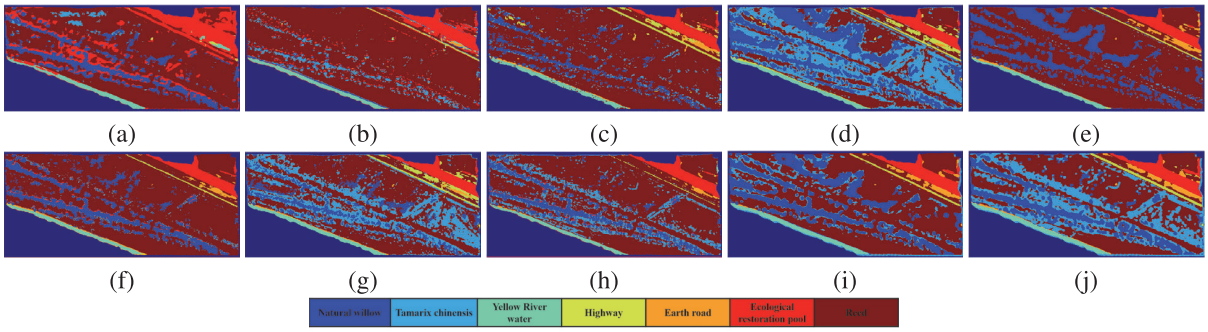


Fig. 8. Data visualization and classification maps for target scene NW-AreaII data obtained with different methods. (a) GAHT (76.88%). (b) SACP (70.54%). (c) CLDA (79.44%). (d) SCLUDA (78.64%). (e) CACL (79.91%). (f) S4DL (81.60%). (g) FSSG (84.71%). (h) SDEnet 87.06%). (i) A2XP (81.36%). (j) SUPnet (92.96%).

TABLE IX

QUANTITATIVE RESULTS OF THE CORRELATION BETWEEN ENDMEMBER VARIABILITY AND SPECTRAL SHIFT ON HOUSTON DATASET

Class		(EM-SD,EM-ED)	(SD,ED)	(SD,TD)	$\Delta_{(SD,TD)-(SD,ED)}$
SAD↓	1	0.0441	0.4104	0.5277	0.1173
	2	0.0424	0.2878	0.1939	-0.0939
	3	0.0431	0.4182	0.2502	-0.1680
	4	0.0419	0.3655	0.4199	0.0544
	5	0.0429	0.1891	0.1347	-0.0544
	6	0.0426	0.1906	0.1144	-0.0762
	7	0.0427	0.1801	0.1249	-0.0552
mean SAD↓		0.0428	0.2916	0.2522	-0.0394

TABLE X

QUANTITATIVE RESULTS OF THE CORRELATION BETWEEN ENDMEMBER VARIABILITY AND SPECTRAL SHIFT ON THE GID DATASET

Class		(EM-SD,EM-ED)	(SD,ED)	(SD,TD)	$\Delta_{(SD,TD)-(SD,ED)}$
SAD↓	1	0.0444	1.3279	1.5322	0.2043
	2	0.0445	0.9412	1.4897	0.5485
	3	0.0443	0.4203	0.6872	0.2669
	4	0.0438	0.4986	0.2595	-0.2391
	5	0.0441	1.2176	0.8547	-0.3629
mean SAD↓		0.0442	0.8811	0.9647	0.0836

the SD comparable to those observed in the TD. The calculated difference $\Delta_{(SD,TD)-(SD,ED)}$ confirms that the spectral shifts introduced by our ED remain within reasonable bounds. The above results confirm a strong correlation between endmember variability within the SD and cross-domain spectral shifts, and verify that domain expansion via latent perturbation in SUG can generate EDs with reasonable spectral shifts, and the magnitude of these shifts is consistent with real TD scenarios.

2) Qualitative Comparison: Spatial distributions of SD and ED for Houston and GID datasets are visualized in Fig. 9. Fig. 9(a) and (c) displays original sample distributions, where • denotes the SD and × represents the ED samples with numerical class labels. These visualizations confirm domain shifts between corresponding classes [e.g., classes 1 and 4 in Fig. 9(a) and (c)], consistent with the SAD values in Tables IX and X. After feature extraction by SUPnet [see Fig. 9(b) and (d)], the learned feature representations show clearly separated class boundaries with substantially reduced interdomain gaps within each class.

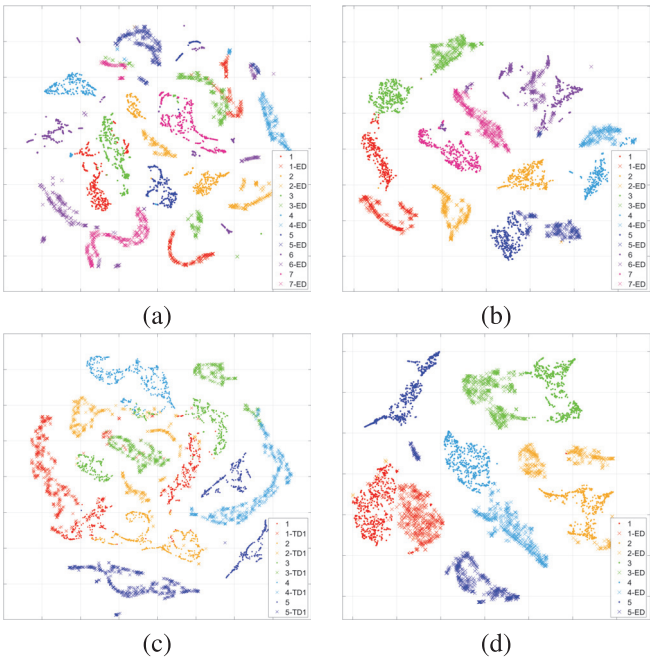


Fig. 9. t-SNE visualization of Houston and GID datasets, where • represents the SD, × represents the ED, and the number represents the class index. (a) Original samples of SD and ED from Houston. (b) SD and ED features of Houston output by SUPnet. (c) Original samples of SD and ED from GID. (d) SD and ED features of GID output by SUPnet. It is obvious that the ED generated by SUPnet has a certain gap with respect to SD, and its extracted features have better class separability.

E. Spectral Unmixing Visualization of SUG

To analyze the effectiveness of the abundance values and endmembers estimated by the SUG in the absence of ground truth, we use Houston dataset as an example. The abundance maps and endmembers output by the trained SUG for the SD are shown in Figs. 10 and 11. In the experiments, the number of endmembers output by the SUG is set to the number of classes in the dataset, which is $n_{em} = 7$ for Houston dataset. The color scale from blue to red in abundance maps indicates the abundance value range from 0 to 1. This range reflects the importance of different endmembers in constituting the spectral signature of each pixel. Notably, in the third abundance map, the tree areas within the scene consistently appear red, indicating that the third endmember significantly contributes to the spectral mixing of the trees. From the local views of the abundance maps, it is observed that the sixth and seventh abundance maps correspond to nonresidential buildings and roads, respectively. In particular, the abundance map for nonresidential buildings is able to demonstrate the

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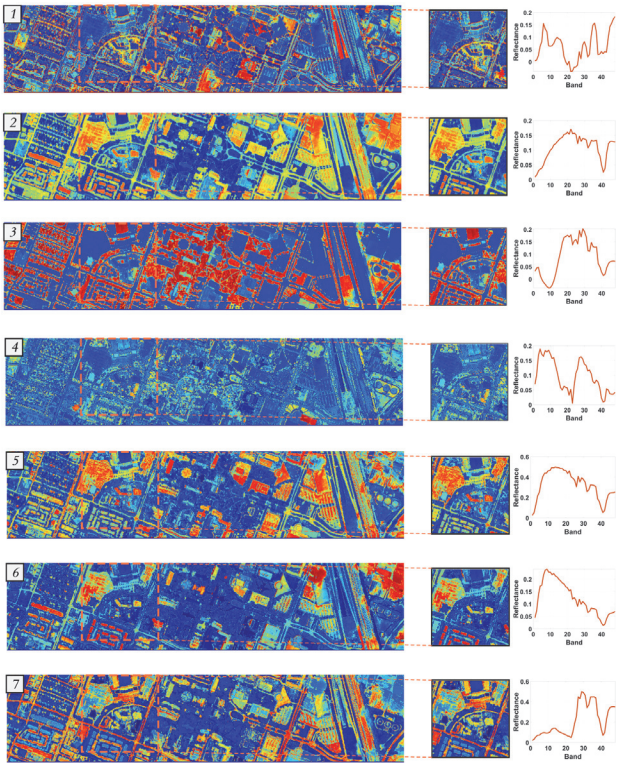


Fig. 10. Abundance maps and endmembers of Houston 2013 dataset estimated by the SUG.

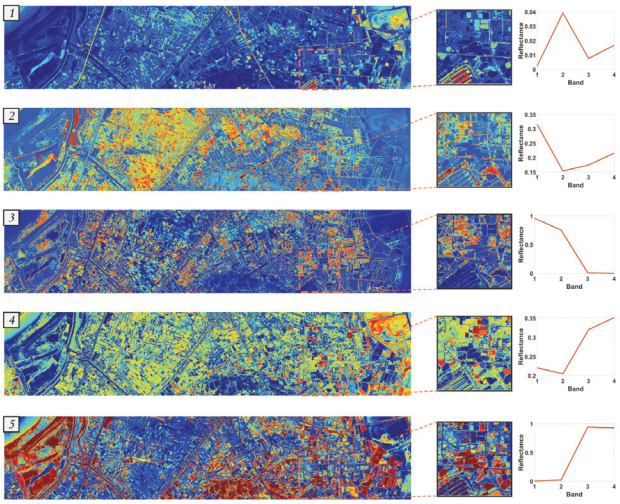


Fig. 11. Abundance maps and endmembers of GID-nc dataset estimated by the SUG.

spectral significance of individual buildings. Therefore, each land cover class has its own dominant endmember reflecting its spectral information.

F. Performance on Spectral Unmixing

To evaluate the performance of SUG in the task of spectral unmixing, we compare four unsupervised spectral unmixing algorithms, including MSNet [34], CyCU-net [35], MUNet [36], and A2SAN [37]. These methods do not require the ground-truth values of abundance values and endmembers during the training phase. SAD and root-mean-square error (RMSE) are used to assess the performance of spectral unmixing.

TABLE XI

QUANTITATIVE RESULT FOR HOUSTON 2013 DATA

Class		Unmixing algorithms				
		MSNet [35]	CyCU-net [36]	MUNet [37]	A2SAN [38]	SUG
SAD↓	1	0.1953	0.8970	0.5989	0.1541	0.0576
	2	0.5748	0.7096	0.2425	0.3296	0.2852
	3	0.5117	0.6738	0.4756	0.5110	0.4799
	4	0.3049	0.7302	0.9097	0.5525	0.5085
	5	0.5360	0.6132	0.7304	0.2070	0.2403
	6	0.4594	0.6569	0.5220	0.3139	0.4178
	7	0.6792	0.8639	0.8685	0.3111	0.2731
mean SAD↓		0.4659	0.7349	0.6211	**0.3399**	0.4232
RMSE↓		0.0694	0.0924	0.0997	0.0621	**0.0498**

TABLE XII

QUANTITATIVE RESULT FOR THE GID-NC DATA

Class		Unmixing algorithms				
		MSNet [35]	CyCU-net [36]	MUNet [37]	A2SAN [38]	SUG
SAD↓	1	1.0411	2.1333	1.1309	0.9697	0.4125
	2	0.8781	0.6611	0.7326	0.9908	0.8576
	3	0.5987	0.3763	1.0391	0.9997	0.1872
	4	1.0678	1.4873	1.0364	1.0138	0.7201
	5	1.2200	2.6136	1.0871	1.0000	0.6918
mean SAD↓		0.9611	1.4543	1.0052	0.9948	**0.5738**
RMSE↓		0.2051	0.4807	0.3004	0.5413	**0.1817**

We compare the unmixing performance of each method on both Houston 2013 data and the GID-nc data. Since cross-domain classification data lacks ground-truth values for abundance and endmembers, we employ a commonly used method in unmixing tasks to construct approximate ground-truth endmembers (by calculating the mean spectrum for each class based on class labels and selecting one spectrum closest to the mean spectrum from all spectra of each class as the approximate ground-truth endmember $\mathbf{M}_i$). Tables XI and XII present the class-specific SAD, mean SAD, and RMSE for each method. The class-specific SAD calculates the SAD for each class between the estimated endmembers $\bar{\mathbf{M}}_i$ and the approximate ground-truth endmembers $\mathbf{M}_i$

$$SAD\left(\bar{\mathbf{M}}_i, \mathbf{M}_i\right) = \arccos\left(\frac{\bar{\mathbf{M}}_i^T \mathbf{M}_i}{\left\|\bar{\mathbf{M}}_i\right\| \left\|\mathbf{M}_i\right\|}\right). \tag{19}$$

The mean SAD is the average of the class-specific SADs. The RMSE is used to measure the similarity between the reconstructed image and the original image.

By comparing the quantitative results in Tables XI and XII, it can be observed that SUG outperforms the other four methods, especially on the GID-nc data where the mean SAD is reduced by 0.42–0.91. Most existing spectral unmixing methods are limited to unmixing small regions and often yield unsatisfactory results when dealing with large-scale scenes. Different from other unsupervised spectral unmixing methods, SUPnet focuses on the classification task for large scenes during the training phase. This allows SUG to indirectly integrate all class information within the scene during the unmixing training, which obviously contributes to enhancing the unmixing performance. In addition, the four comparison methods such as MSNet and CyCU-net are only applicable to HSI data. When processing GID-nc (four-band multispectral data), due to the limited information in the spectral dimension, they struggle to accurately extract the endmembers of various land cover classes. In particular, the spectral differences

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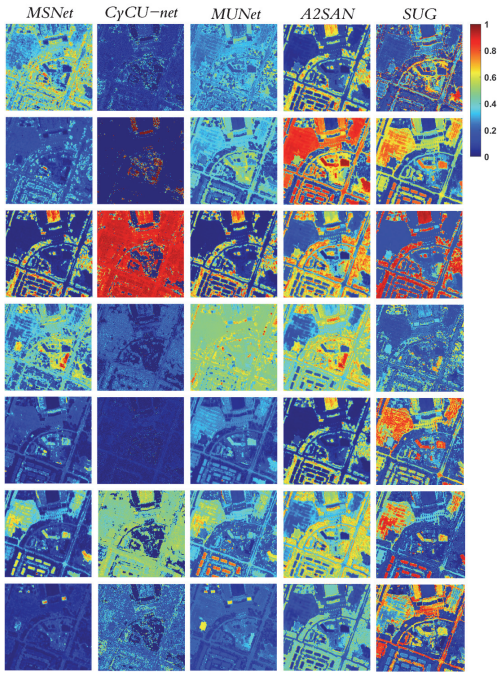


Fig. 12. Abundance maps for Houston 2013 data obtained with different unsupervised spectral unmixing methods.

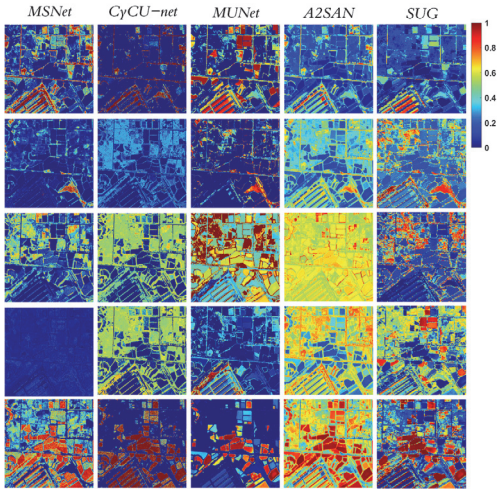


Fig. 13. Abundance maps for GID-nc data obtained with different unsupervised spectral unmixing methods.

between the fourth class (river) and the fifth class (lake) in the GID-nc dataset are small, and the SAD of all comparison methods is greater than 1 in the fourth class and the fifth class.

Figs. 12 and 13 present the visualization results of abundance maps for all methods, with each column representing the abundance maps output by a different method. Referring to Figs. 10 and 11, we show local areas extracted from the global scene. It is evident that SUG can extract more accurate abundance values for most land cover classes. For instance, in the third and sixth rows of Fig. 12, SUG successfully separates the tree areas and nonresidential building areas. CyCU-net, as an unmixing method based on an autoencoder, estimates invalid abundance values for most classes. As evident in the second column of Fig. 12, this method fails in unmixing on a larger scene. From the visualization results, MSNet, MUNet, and A2SAN perform better than CyCU-net in unmixing, but they still cannot effectively separate the abundance values of some land cover classes, such as residential building areas (fifth row of Fig. 12), road areas (seventh row of Fig. 12), and garden land areas (third row of Fig. 13).

IV. Conclusion

In this article, a SUPnet is proposed under a single-DG framework for cross-domain hyperspectral image interpretation. First, considering the potential relationships between endmember variability and changes in environmental conditions across different scenes, a SUG is designed to estimate the abundance values and endmembers of the SD while implementing a spectral unmixing-guided domain expansion strategy. This strategy is suitable for spectral images and has more physical significance. Furthermore, inspired by the transfer paradigm of multidomain-specific representations, an MPL approach is developed. By integrating the data and endmembers of ED output from SUG, prompt learning is implemented at both the domain level and the endmember level to capture prompts with data knowledge and endmember knowledge. In addition, domain and endmember adapters are designed to perceive the correlations of multilevel domain-specific knowledge across different domains, continuously updating the adapted prompts to make it as adaptable as possible to predicting the TD. Extensive experimental results demonstrate that the proposed SUPnet can effectively estimate data abundance values and endmembers, and outperform some state-of-the-art approaches on multiple cross-domain classification tasks.

Despite the promising results achieved by SUPnet, several important avenues remain for further exploration. First, we plan to extend the current DG paradigm to open-world scenarios, where the TD may contain novel land cover categories not observed during training. This necessitates the ability to not only handle spectral shifts but also to detect and adapt to emerging semantic classes in an online manner. Second, we intend to incorporate foundation models (e.g., vision-language models and large-scale pretrained Transformers) into the cross-scene classification framework. Leveraging their strong representation power and semantic prior knowledge, foundation models offer a promising pathway to further enhance generalization across diverse and unseen scenes. We believe these two directions will collectively push the boundaries of cross-domain HSI interpretation toward more practical and intelligent applications.

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